Galaxy for analysis of human health data

Wolfgang Maier
Galaxy Europe Team, Freiburg

Galaxy
PROJECT

Galaxy is a web-based platform for giving users graphical, OS-independent, controlled access to attached compute and storage for performing any kind of data analysis

Data intensive science for everyone.

galaxyproject.org
View license
Code of conduct
Contributing
Security policy
1.7k stars **1.1k** forks **64** watching **61** Branches **237** Tags Activity
Custom properties
Public repository

dev | Go to file | + | Code

jmchilton Merge pull request #21490 from davelopez/... 981b5f5 · yesterday

.ci	Drop unneeded postgres and rabbit...	4 months ago
.circleci	Drop support for Python 3.8	11 months ago
.github	add client-api package to automate...	last week
.vscode	WIP: modern tool shed frontend	3 years ago
client-api	fix sync-version.js to handle patch ve...	last week
client	Merge pull request #21490 from dav...	yesterday
config	Remove trackster entry point mako	3 months ago

Releases 33

v25.1.0 Latest
last month

+ 32 releases

Packages 1

galaxy

Contributors 363

+ 349 contributors

Languages

Python 63.5% **TypeScript** 12.8% **Vue** 12.1% **JavaScript** 4.3%
Jupyter Notebook 3.2% **Roff** 1.4% **Other** 2.7%

Galaxy Europe

Using 0 b | Login | Register

Upload
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Tools

Discover Tools

search tools

Get Data
Send Data
Collection Operations

GENERAL TEXT TOOLS

Text Manipulation
Convert Formats
Filter and Sort
Join, Subtract and Group

Column join on multiple datasets

Compare two Datasets to find common or distinct rows

Datamash (operations on tabular data)

Group data by a column and perform aggregate operation on other columns.

Join two Datasets side by side on a specified field

Reverse columns in a tabular file

Subtract the intervals of two datasets

Transpose rows/columns in a tabular file

Learn Galaxy with Galaxy Training Network

"Anyone, anywhere in the world should have free, unhindered access to not just my research, but to the research of every great and enquiring mind across the spectrum of human understanding." – Prof. Stephen Hawking

News

GPU-accelerated genome mapper Parabricks FQ2BAM is now in Galaxy

NVIDIA Parabricks GPU-accelerated FQ2BAM is now available in Galaxy, bringing FASTQ-to-BAM alignment runtime to just a few minutes for 2.5 GB paired-end human whole-genome datasets. Its

Events

Jan 9 - Jan 14 **PAG 33 Plant and Animal Genome Conference**

Join PAG 33 to explore breakthroughs in plant and animal genomics and bioinformatics.

Jan 14 **Galaxy for analysis of human health data**

History

search datasets

Unnamed history

Default Storage

0 B

This history is empty.

You can load your own data or **get data from an external source.**

bcftools

Search All Installed Only

9838 repositories available at **https://toolshed.g2.bx.psu.edu/**

Name	Owner	Downloaded	Updated
bcftools_consensus Wrapper for bcftools application bcftools consensus.	iuc	>3k	today
bcftools_annotate Wrapper for bcftools application bcftools annotate.	iuc	>3k	today
bcftools_call Wrapper for bcftools application bcftools call.	iuc	>2k	today
bcftools_convert_from_vcf Wrapper for bcftools application bcftools convert from vcf.	iuc	>2k	today
bcftools_filter Wrapper for bcftools application bcftools filter.	iuc	>2k	today
bcftools_view Wrapper for bcftools application bcftools view.	iuc	>2k	today
bcftools_mpileup Wrapper for bcftools application bcftools mpileup.	iuc	>2k	today
bcftools_norm Wrapper for bcftools application bcftools norm.	iuc	>2k	today

BCFtools are meant as a faster replacement for most of the perl VCFtools commands.

Show additional details and dependencies.

Revision	Tools and Versions		Requires	Tests	Status	
24	bcftools_norm	1.22+galaxy0	+24.1	✓	Installed	Uninstall

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Tools

Discover Tools

bcftools norm

Show Sections

bcftools norm Left-align and normalize indels; check if REF alleles match the reference; split multiallelic sites into multiple rows; recover multiallelics from multiple rows

bcftools norm
Left-align and normalize indels; check if REF alleles match the reference; split multiallelic sites into multiple rows; recover multiallelics from multiple rows
(Galaxy Version 1.22+galaxy0)

Run Tool

- Selected 1.22+galaxy0
- Switch to 1.15.1+galaxy4
- Switch to 1.15.1+galaxy3
- Switch to 1.15.1+galaxy2
- Switch to 1.15.1+galaxy0
- Switch to 1.10
- Switch to 1.9+galaxy2
- Switch to 1.9+galaxy1
- Switch to 1.9
- Switch to 1.4.0.0

Tool Parameters

VCF/BCF Data *

1: unprocessed called variants

accepted formats

Choose the source for the reference genome

Use a built-in genome

Reference genome *

Human Dec. 2013 (GRCh38/hg38) (hg38)

When any REF allele does not match the reference genome base *

- (•) ignore the problem (-w)
- ( ) exclude the variant record from the output (-wx)
- ( ) fix the variant record using the reference genome information (-ws)
- ( ) exit with an error (-e)

Warnings about REF mismatches will be emitted to the standard error (stderr) stream, and it is recommended to check there for problems if you choose not to exit with an error immediately upon encountering a mismatch.

Atomize *

- (•) Do not atomize
- ( ) Atomize and use * ALT allele for proper genotype representation (--atomize)
- ( ) Atomize and accept missing genotype information (legacy behavior; --atomize --atom-overlaps .)

Decompose complex variants (e.g. MNVs become consecutive SNVs)

Left-align and normalize indels?

No

History

search datasets

Unnamed history

Default Storage

2.65 kB

1: unprocessed called variants

Galaxy Training!

Welcome to Galaxy Training!

Collection of tutorials developed and maintained by the worldwide community

Galaxy for Scientists

We have separated the tutorials into several categories based on field and technology. W
exploring other ways to organise the tutorials going forward!

Start Here

Topic	Tutorial
Introduction to Galaxy Analyses	14
Using Galaxy and Managing your Data	27

Not sure where to start?

Try the NGS Basics Learning Path!

Start

Scientific Fields

Topic	Tutorials
Climate	14
Computational chemistry	9

Methodologies

Contributor Hall of Fame

511	35	498	10.5
Contributors	Topics	Tutorials	Years

This project would not be possible without the many amazing community contributors!

View the Hall of Fame | View all GTN stats

Do you want to help with this project and join our Hall of Fame? Please see our dedicated tutorials or our Frequently Asked Questions to get you started.

Acknowledgment and Funding

More information about this project can be found in our publication in PLOS Computational Biology. We would like to thank all contributors to our Galaxy training materials, the Galaxy community for their constant support, and our funding sources:

AI4Life, ACME, BioNT, BeYond-COVID, CINECA, ELIXIR-CONVERGE, EDCTP-EU, DASH UK, EOSC-Life, EPSRC/OU, EuroScienceGateway, Fair-Ease, FNSO, Gallantries, DEFEND, NFDI4Bioimage, NFDI4Culture, DataPLANT

CUT&RUN data analysis

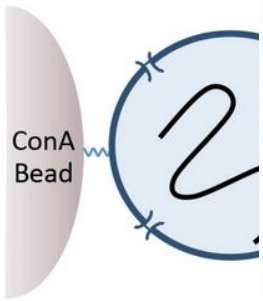
Author(s) Florian Heyl, Pavankumar Videm

Editor(s) Helena Rasche, Lucille Delisle, Wolfgang Maier

Reviewers

When working with real data
Step 1: Preprocessing
Step 2: Mapping
Step 3: Filtering Mapped Reads and second level Quality Control
Step 4: Peak calling
Step 5: Identifying Binding Motifs
Step 6: Visualisation of Coverage
Conclusion
Frequently Asked Questions
References
Feedback
Citing this Tutorial

In many organism, transcription factors (TF)
2,800 proteins and more than 1,600 are TF
the role of TFs, such as GATA1, is a very imp
source of a disease, such as myelofibrosis a



When working with real data
Step 1: Preprocessing
Get Data
Quality Control
Trimming Reads
Step 2: Mapping
Step 3: Filtering Mapped Reads and second level Quality Control
Step 4: Peak calling
Step 5: Identifying Binding Motifs
Step 6: Visualisation of Coverage
Conclusion
Frequently Asked Questions
References
Feedback
Citing this Tutorial

Quality Control

We first have to check if our data contains adapter sequences that we have to remove. A typical CUT&RUN experiment has a read length of 30-80 nt. We can check the raw data quality with **Falco**.

Hands On: Quality Control

1. **Flatten collection** with the following parameters convert the list of pairs into a simple list:
   - *"Input Collection"*: `2 PE fastqs`
2. **Falco** (Galaxy version 1.2.4+galaxy0) with the following parameters:
   - *"Raw read data from your current history"*: Choose the output of **Flatten collection** selected as **Dataset collection**.
3. Inspect the web page output of **Falco** for the `Rep1_forward` sample. Check what adapters are found at the end of the reads.

Question

1. How many reads are in the FASTQ?
2. Which sections have a warning or an error?

Solution

1. There are 300,000 reads.
2. The 4 steps below have warnings:
   1. **Per base sequence content**

Galaxy platform + Compute/Storage Infrastructure + Tools + Reference Data + Training material

Resource	Release	Tier	Summary	Tools	References	Region
UseGalaxy.eu	25.0		The free public European Galaxy server	3,844	1,336 genomes	DE
UseGalaxy.fr	25.0		General purpose genomics Galaxy server	2,118	1,214 genomes	FR
UseGalaxy.org (Main)	25.1		The Galaxy Project free public server; biomedical research	2,110	4,503 genomes	US
UseGalaxy.org.au	25.0		The Australian UseGalaxy server.	2,196	1,232 genomes	AU

Recent developments around Galaxy

Galaxy
PROJECT

Workflows & the IWC workflow library

Galaxy IWC - Workflow Library

About Contribute GitHub

Open, peer-reviewed workflows for reproducible science

Get started with one of our most popular workflows, or browse the full library below.

RNA-Seq Analysis: Paired-End Read Processing and Quantification

Complete RNA-Seq analysis for paired-end data: Processes raw FASTQ data through adapter and bad quality removal (fastp), alignment with STAR using...

Transcriptomics

1.2 Jan 27, 2025

ChIP-seq Analysis: Paired-End Read Processing

Complete ChIP-seq analysis for paired-end sequencing data. Processes raw FASTQ files through adapter removal (fastp), alignment to reference...

Epigenetics

1.0 Oct 3, 2025

ATAC-seq Analysis: Chromatin Accessibility Profiling

Complete ATAC-seq analysis pipeline for paired-end reads. Processes raw FASTQ data through adapter and bad quality removal (cutadapt), alignment (Bowtie2...

Epigenetics

1.0 Nov 28, 2024

CATEGORIES

- Microbiome 16
- Vertebrate Genome Pr... 13
- Epigenetics 12
- Transcriptomics 11
- Variant Calling 10
- Metabarcoding 9
- Single Cell 8
- Computational Chemist... 7
- Genome Annotation 7
- SARS-COV-2 7

Search workflows...

109 workflows

List Grid

Scaffolding with Hi-C data VGP8

This workflow performs the scaffolding of a genome assembly using HiC data with YAHS. Can be used on any assembly with Hi-C data, and the assembly in the gfa format.

Vertebrate Genome Project

3.3

Dec 10, 2025

Metagenomic Genes Catalogue Analysis

Metagenomic analysis, from raw reads to gene catalog. Uses Megahit to assemble contigs and Prodgial to predict CDSs on contigs to provide the gene catalog. Finally, functional, taxonomic, and antimicrobial resistance information is provided.

Genome Annotation Metagenomics Microbiome

1.1

Dec 8, 2025

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Show all workflow runs
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Invoked Workflow: Variant calling and consensus construction from paired-end Illumina data (WGS or tiled-amplicon) of simple, non-segmented viral genomes (imported from uploaded file) (Version: 6)

Run Rerun

invoked 5 months ago

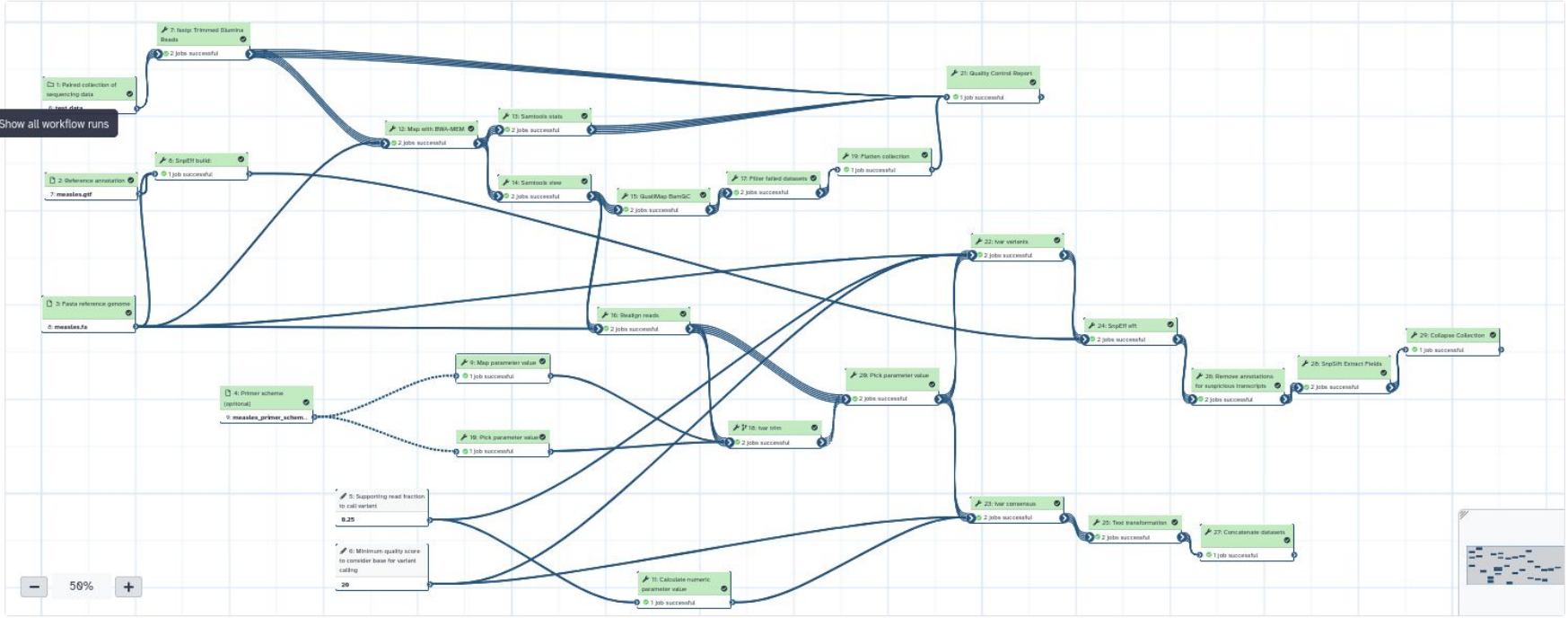
History: Variant calling and consensus construction fro...

29 of 29 steps successfully scheduled.

38 of 38 jobs complete.

Peter van Heusden | Wolfgang Maier | workflow runs: 2

Overview | Steps | Inputs | Outputs | Report | Export | Metrics



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Overview | Steps | Inputs | Outputs | Report

Show all workflow runs

50%

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Overview | Steps | Inputs | Outputs | Report | Export | Metrics

Export

Invocation Export Wizard

1 Select output format
2 Select destination
3 Export

Select the format you would like to export the workflow invocation to and click Next to continue.

Research Object Crate

Workflow Run RO-Crate

Learn more

BioCompute Object

Learn more

Compressed File

Export the invocation to a compressed File containing the invocation data in Galaxy native format.

Next

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Invoked Workflow: Variant calling and consensus construction from paired-end Illumina data (WGS or tiled-amplicon) of simple, non-segmented viral genomes (imported from uploaded file) (Version: 6)

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Overview | Steps | Inputs | Outputs | Report

Show all workflow runs

Export

Invocation Export Wizard

1 Select output format

2 Select destination

3 Export

Metrics

Timing: Seconds | Group By: Tool ID

Aggregate Runtime Time (in Seconds)

Runtime (in Seconds)

Tool ID

collapse_dataset/5.1.0
calculate_numeric_param/0.1.0
lofreq_viterbi/2.1.5+galaxy0
multiqc/1.24.1+galaxy0
snpEff/5.2+galaxy0
pick_value/0.2.0
qualimap_bamqc/2.3+galaxy0
snpEff_build_gb/5.2+galaxy0
fastp/1.0.1+galaxy0
tp_find_and_replace/9.5+galaxy2
tp_sed_tool/9.5+galaxy2
ivar_consensus/1.4.4+galaxy0
ivar_variants/1.4.4+galaxy0
samtools_stats/2.0.7
snpSift_extractFields/4.3+t.gal...
samtools_view/1.21+galaxy0
map_param_value/0.2.0
ivar_trim/1.4.4+galaxy1

Aggregate Allocated Core Time (in Seconds)

Allocated Core Time (in Seconds)

Tool ID

qualimap_bamqc/2.3+galaxy0
collapse_dataset/5.1.0
lofreq_viterbi/2.1.5+galaxy0
fastp/1.0.1+galaxy0
ivar_trim/1.4.4+galaxy1
bwa_mem/0.7.19
ivar_consensus/1.4.4+galaxy0
ivar_variants/1.4.4+galaxy0
snpSift_extractFields/4.3+t.gal...
samtools_view/1.21+galaxy0

Tool ID toolshed.g2.bx.psu.edu/repos/devteam/bwa/bwa_mem/0.7.19

Allocated Core Time (in Seconds) 128

File

ised File containing the native format.

Next

UseGalaxy EU users, jobs, and related

Tools	Histories	Registered users
4160	10,292,774	153,214

Jobs	Datasets	Galaxy Elixir AAI Users
95,250,788	187,601,870	336

Workflow executions
661,817

Galaxy monthly active users

Month	Active users
2025-12	7930

Utilization of Galaxy Europe's compute resources

Total CPU cores	Claimed CPU cores	Unclaimed CPU co...	Total mem (TiB)	Claimed mem (TiB)	Unclaimed mem (T...
7348	6669	677	42.0	39.7	2.21

Interactive data exploration
(single-cell data example)

Galaxy Single Cell Omics

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SAM/BAM
SINGLE CELL TOOLS
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RaceID
Monocle3
Multiomics
Inference
Spatial
Seurat
SC3
HCA-Scanpy
Scater
SCCAF
HiCExplorer
Graph/Display Data
Interactive tools
AUTOMATED CELL ANNOTATION
SCMap
ScPred
Garnett

SINGLE CELL LAB

Home
Learn to use Galaxy for single-cell analysis
Workflows
Community

Welcome to the Galaxy Europe Single Cell Lab!
Access Single Cell Omics analysis tutorials, tools, and workflows that let you a … or private infrastructure.
How does this page relate to Galaxy Europe?

Learn to use Galaxy for single-cell analysis

Learning pathways | Importing public data | Converting datatypes | Visualising Data | Help

Using public data is free! Learn how to retrieve data from common sources.

- Importing files from the Single Cell Expression Atlas
- Importing from public atlases - Tutorial
- Importing from NCBI/GEO
- Importing 10X Files
- `Seurat Read10x`
- `DropletUtils Read10x`

DropletUtils provides a number of utility functions for handling single-cell (RNA-seq) data from droplet technologies such as 10X Genomics. This includes data loading, identification of cells from empty droplets, removal of barcode-swapped pseudo-cells, and downsampling of the count matrix.

https://singlecell.usegalaxy.eu/

an outstanding example of community effort and a Galaxy Labs page.

Galaxy Training! | Learning Pathways | Help | Settings | Search Tutorials

Galaxy Community Building

Material

Special Interest Groups & Communities

Lesson	Slides	Hands-on
Creating community content		
Creation of a Galaxy tutorial table for your community (Community, SIG)		
Creation of resources listing Galaxy workflow for your community (Community, SIG)		
Creation of resources listing all the tools and their metadata relevant to your community (Community, SIG, CoDex)		
Creation of the labs in the different Galaxy Instances for your community (Community, SIG, CoDex)		
What's a Special Interest Group?		
Creating a Special Interest Group		

Subdomains & Galaxy Labs

Lesson	Slides	Hands-on	Reco
Make your tools available on your subdomain (subdomain, earth-system, Interactive-tools)			

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Join, Subtract and
Convert Formats
FASTA/FASTQ
SAM/BAM
SINGLE CELL TOOLS
Single-cell
Import/manipulat
Scanpy
RaceID
Monocle3
Multiomics
Inference
Spatial
Seurat
SC3
HCA-Scanpy
Scater

SINGLE CELL LAB

Welcome to the Galaxy Europe Single Cell Lab!

thout programming experience or

Visualising Data

Further training

Workflows

39: Annotated_Object

Input
X
layers
obs
Barcode
_index
batch
cell_type
emptyFDR
emptyLimited
emptyLogProb

layers
Display Inspect

Group

Path	/layers
Name	layers
Raw	▸ Inspect

Attributes

encoding-type	"dict"
encoding-version	"0.1.0"

History

search datasets

Copy of 'LOCKED | Filter, Plot and Explore Single-cell RNA-seq Data | Answer Key'

Aligns with this training: https://training.galaxyproject.org/training-material/...

#single-cell #training

Default Storage

1.64 GB 43 13 1

51: Interactive CELLxGENE VIP Environment on dataset 39

#n701-wt-male #n702-ko-male #n703-ko-female #n704-wt-male #n705-wt-male #n706-wt-male #n707-ko-male

40: Annotated_Plots

#n701-wt-male #n702-ko-male #n703-ko-female #n704-wt-male #n705-wt-male #n706-wt-male #n707-ko-male

39: Annotated_Object

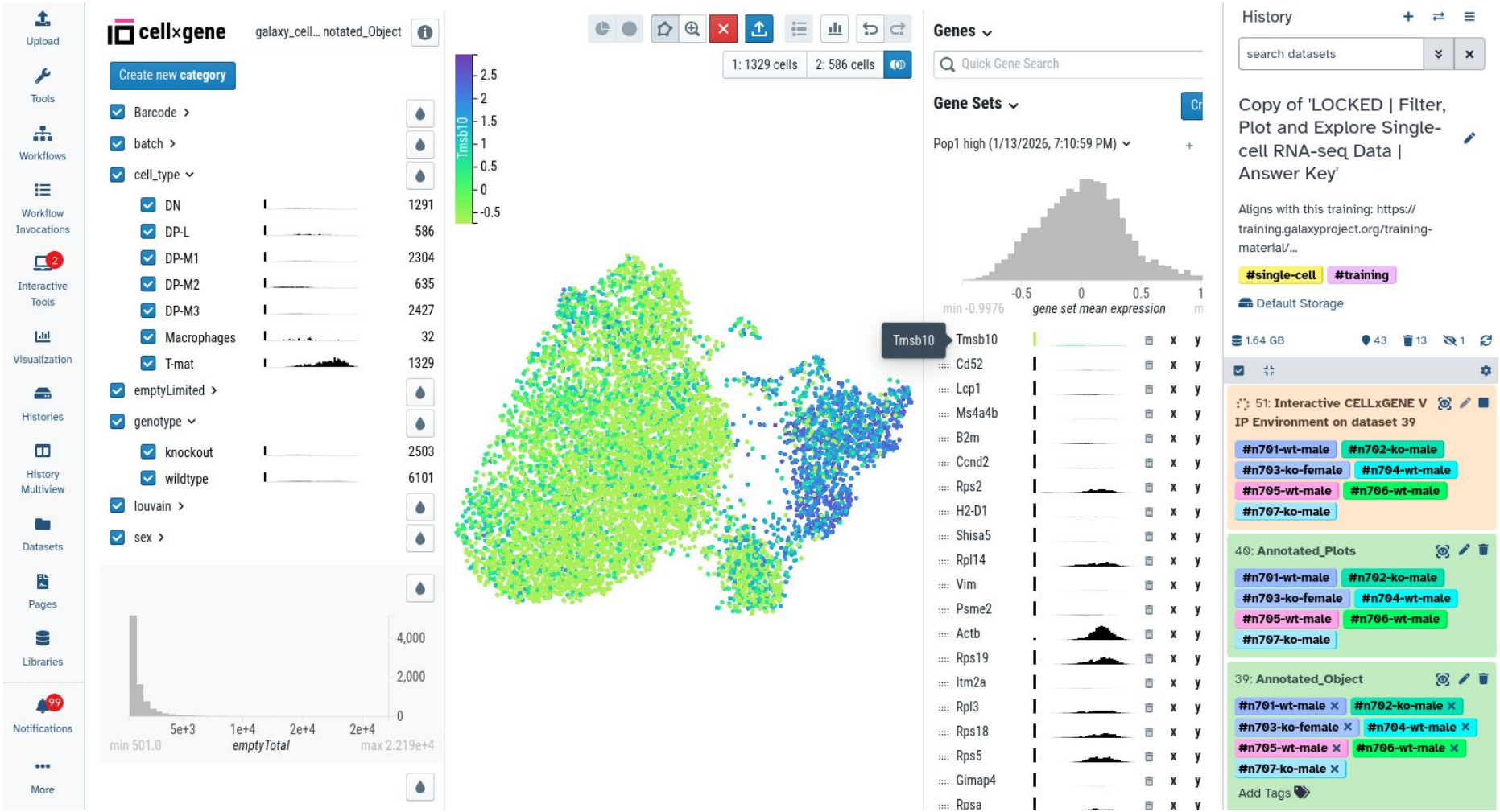
View

#n701-wt-male ✕ #n702-ko-male ✕ #n703-ko-female ✕ #n704-wt-male ✕ #n705-wt-male ✕ #n706-wt-male ✕ #n707-ko-male ✕

Add Tags

Anndata file (129.8 MB)

format **h5ad**, database ?



based on https://github.com/chanzuckerberg/cellxgene

Upload from Disk or Web to **Copy of 'LOCKED | Filter, Plot and Explore Single-cell RNA-seq Data | Answer Key'**

Regular | Composite | Collection | Rule-based

Upload
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Drop files her

Type (set all): Auto-detect | Reference

Choose local file | Choose from repository

Jan 14 Galaxy f | Uplo

search

Label	Details
TF nextcloud	-
My Zenodo	-
FTP Directory	Galaxy User's FTP Directory
ERGA Nextcloud	ERGA Pilot storage provided by B
Hugging Face Hub 🤗	Hugging Face Hub is a platform f datasets.
Tabula Atlases	Tabula Sapiens and Tabula Muris providing comprehensive cell-typ multiple tissues and organs in hu
COVID-19 FASTQ	COVID-19 RAW sequences in FAS
PDB	Protein Data Bank (PDB)
ML models	ML/AI models for PyTorch, Tenso
Zenodo	Zenodo is an open-access digital Organization for Nuclear Researc cite research outputs across vari
EBI FTP server	European Bioinformatic Institute
SARS-CoV-2 result files	SARS-CoV-2 variants and consen

+ Create new

« ‹ 1 2 ›

search

Label	Time
Testis_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:49 PM
Eye_TSP1_30_version2d_10X_smartseq_scvi_Nov122024_updated.h5ad	Oct 14, 2025, 6:43 PM
Lung_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:45 PM
Bone_Marrow_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:42 PM
Pancreas_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:47 PM
Small_Intestine_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:48 PM
Blood_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:42 PM
Tongue_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:50 PM
Uterus_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:50 PM
Stomach_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:49 PM
Bladder_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:42 PM
Lymph_Node_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:46 PM
Salivary_Gland_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:48 PM
Kidney_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:44 PM
Ovary_TSP1_30_version2d_10X_smartseq_scvi_Nov262024.h5ad	Oct 14, 2025, 6:47 PM
Prostate_TSP1_30_version2d_10X_smartseq_scvi_Nov122024.h5ad	Oct 14, 2025, 6:47 PM

‹ Back | « ‹ 1 2 › » | ✕ Cancel | ✓ Select

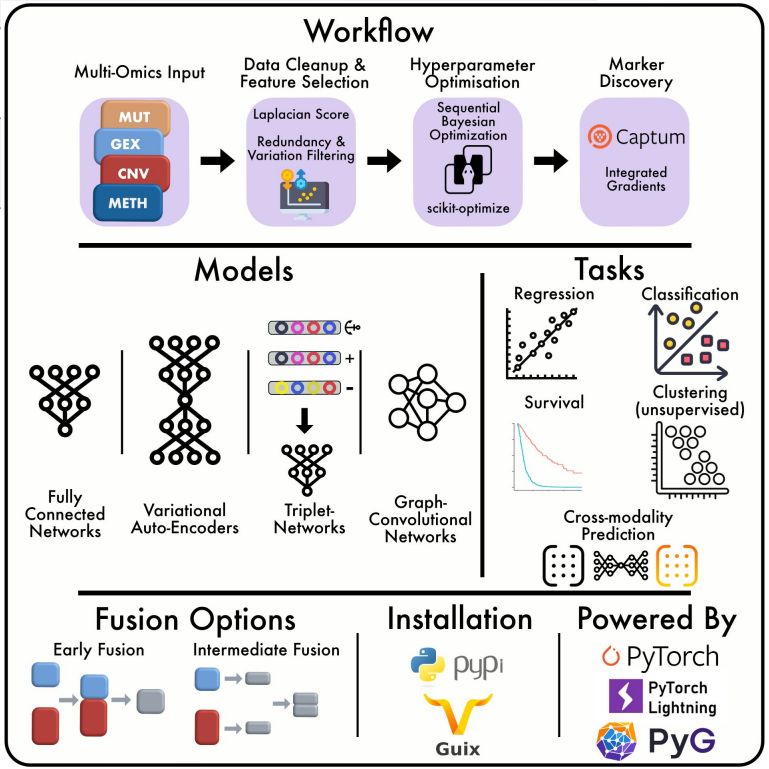
https://tabula-sapiens.sf.czbiohub.org/

Machine learning

Flexynesis - multi-omics Integration

Tutorials covering the integration of multi-omics data using advanced statistical and machine learning techniques with Flexynesis package.

Lesson	Slides	Hands-on	Recordings	Input dataset	Workflows
Identifing Survival Markers of Brain tumor with Flexynesis					
Modeling Breast Cancer Subtypes with Flexynesis					
Prepare Data from CbioPortal for Flexynesis Integration					
Unsupervised Analysis of Bone Marrow Cells with Flexynesis					



Article | Open access | Published: 12 September 2025

Flexynesis: A deep learning toolkit for bulk multi-omics data integration for precision oncology and beyond

Bora Uyar, Taras Savchyn, Amirhossein Naghsh Nilchi, Ahmet Sarigun, Ricardo Wurmus, Mohammed Maqsood Shaik, Björn Grüning, Vedran Franke & Altuna Akalin

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flexynesis

Show Sections

Flexynesis A deep-learning based multi-omics bulk sequencing data integration suite

Flexynesis cBioPortal import and prepare cBioPortal data for Flexynesis analysis

Flexynesis plot tool for visualizing flexynesis results

Flexynesis utils Utility functions for Flexynesis

Flexynesis A deep-learning based multi-omics bulk sequencing data integration suite
(Galaxy Version 0.2.20+galaxy3)

Run Tool

Tool Parameters

I certify that I am not using this tool for commercial purposes.

Yes

Type of Analysis

Supervised training

Training clinical data *

2: train_clin_lgggbm.tabular

accepted formats

Test clinical data *

1: test_clin_lgggbm.tabular

History: Survival Markers of Lower Grade Glio...

Flexynesis on data 3, data 4, and others: plots

a list with 2 **jpg** datasets

Download | Show Details | Run Job Again

1: **clin_km_km_risk_subtypes**

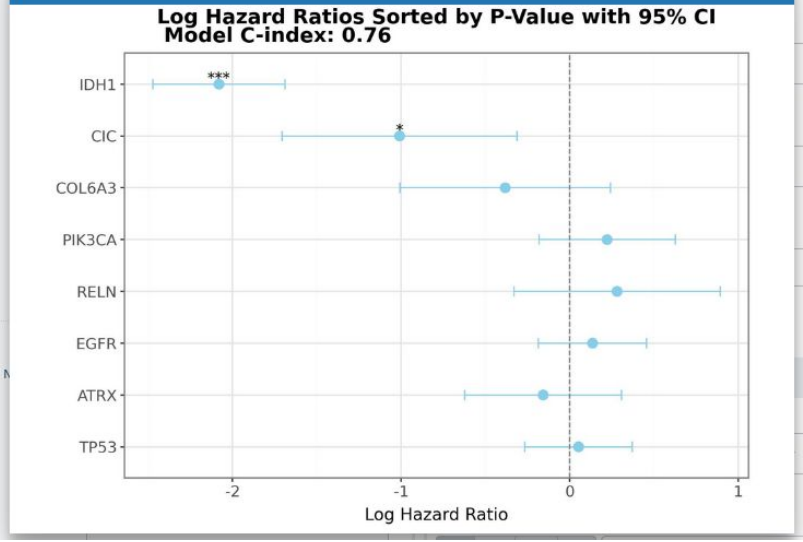
2: **job.feature_importance.IntegratedGradients_cox_hazard_ratios**

133.6 KB

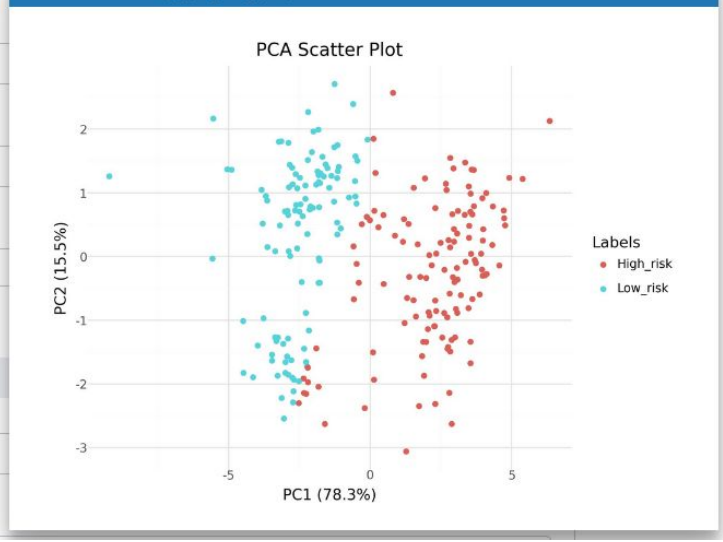
format **jpg**, database **?**

Image in jpg format

undefined: job.feature_importance.IntegratedGradients_cox_hazard_ratios



undefined: embeddings_pca_risk_groups



File sources and storage

Galaxy Europe

Upload
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User Preferences

Manage your user preferences on this page, including email address, password, and other settings.

Manage Information
Edit your email, addresses and custom parameters or change your public name.

Change Password
Allows you to change your login credentials.

Manage Toolbox Filters
Customize your Toolbox by displaying or omitting sets of Tools.

Set Dataset Permissions for New Histories
Grant others default access to newly created histories. Changes made here will only affect histories created after these settings have been stored.

Manage Galaxy API Key
Access your current Galaxy API key or create a new one.

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Manage your tools credentials groups for accessing external services.

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Delete Account
Click here to delete your account.

Sign Out of All Sessions
Click here to sign out of all sessions.

User Preferences / My Repositories / Create New

Onedata

If you have an Onedata account, you can use this repository to import and export your data directly from and to Onedata.
The minimal supported Onezone version is 21.02.4. More information on Onedata can be found on Onedata's website.

Select

Amazon Web Services Private Bucket

Setup access to a private AWS bucket using a secret access key.

Select

Amazon Web Services Public Bucket

Setup anonymous access to a public AWS bucket.

Select

Azure Blob Storage

Setup access to your Azure Blob Storage.

Select

B2DROP by BSC

B2DROP is a Nextcloud to sync and share your research data provided by BSC.

Select

B2DROP by EUDAT

B2DROP is a Nextcloud to sync and share your research data provided by EUDAT.

Select

Dataverse

The Dataverse Project is an open-source web application for sharing, preserving, citing, exploring, and analyzing research data. It enables institutions and researchers to host and manage datasets with rich metadata and persistent identifiers. This configuration allows you to connect Galaxy to a Dataverse instance of your choice.
Note: Your **registered email address** will be used to create datasets in Dataverse.

Select

Dropbox

Connect to your Dropbox account to download and upload files.

Select

eLabFTW

eLabFTW is a free and open source electronic lab notebook from Deltablot. It can keep track of experiments, equipment, and materials from a research lab. Each lab can either host their own installation or go for Deltablot's hosted solution. Here you can connect to an eLabFTW instance of your choice.

Select

FTP Server

Setup connections to FTP and FTPS servers to download and upload files.

Select

Google Drive

Configure your Google Drive account to accept file uploads. Only files uploaded by this Galaxy instance can be read.

Select

Hugging Face Hub

Hugging Face Hub is a platform for sharing machine learning models and datasets. Using this plugin you can connect to the Hugging Face Hub to access public or private machine learning models repositories.

Select

InvenioRDM

InvenioRDM is a research data management platform that allows you to store, share, and publish research data. It is built on top of the Invenio framework and is designed to be highly customizable. Here, you can connect to an InvenioRDM instance of your choice.

Select

OMERO

Connect to an **OMERO** server to browse and import images stored there. You can connect to the Image Data Resource (IDR) or to your own OMERO server instance. When connecting to **IDR**, you can use the public credentials (username: "**public**", password: "**public**").

Select

RSpace

RSpace is an electronic lab notebook (ELN) designed for documenting, managing and sharing scientific data. This template configuration allows you to connect to an RSpace instance of your choice.

Select

S3 Compatible repositories with Credentials

The APIs used to connect to Amazon's S3 (Simple Storage Service) have become something of an unofficial standard for cloud storage across a variety of vendors and services. Many vendors offer storage APIs compatible with S3. Here, you can configure any S3 based repository as long as you have the relevant credentials.

WebDAV (e.g. Nextcloud, Owncloud, Opencloud)

The WebDAV protocol is a simple way to access files over the internet. Here, you can connect to any WebDAV server.

Zenodo

Zenodo is an open-access repository for research data, software, publications, and other digital artifacts. It is developed and maintained by CERN and funded by the European Commission as part of the OpenAIRE project. Zenodo provides a free platform for researchers to share and preserve their work, ensuring long-term access and reproducibility. Zenodo is widely used by researchers, institutions, and

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Amazon Web Services Public Bucket

Setup anonymous access to a public AWS bucket.

Azure Blob Storage

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Select

B2DROP

B2DROP i
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Dropbo

Connect t

FTP Server

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Select

Google

Configure
uploaded

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Select

OMERO

Connect t
there. You
own OME
public cre

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WebDA

The WebD
Here, you

Using 27.6 TB

wolfgang

Upload from Disk or Web to **Survival Markers of Lower Grade Gliomas**

Regular | Composite | Collection | Rule-based

Drop files here

Type (set all): Auto-detect

Reference (set al

Choose local file | Choose from repository | Pa

ver. This should be the username you use to log in to the WebDAV server.

ver. This should be the password you use to log in to the WebDAV server.

History

search datasets

Survival Markers of Lower Grade Gliomas

search

Label	Details
TF nextcloud	-
My Zenodo	-
FTP Directory	Galaxy User's FTP Directory
ERGA Nextcloud	ERGA Pilot storage provided by BSC
Hugging Face Hub 🤗	Hugging Face Hub is a platform for sharing machine learning models and datasets.
Tabula Atlases	Tabula Sapiens and Tabula Muris senis single-cell transcriptomic atlases, providing comprehensive cell-type-resolved gene expression profiles across multiple tissues and organs in human and mouse.
COVID-19 FASTQ	COVID-19 RAW sequences in FASTQ format
PDB	Protein Data Bank (PDB)
ML models	ML/AI models for PyTorch, TensorFlow, Keras ...
Zenodo	Zenodo is an open-access digital repository created by CERN (the European Organization for Nuclear Research) to enable researchers to share, preserve, and cite research outputs across various disciplines.
EBI FTP server	European Bioinformatic Institute FTP server
SARS-CoV-2 result files	SARS-CoV-2 variants and consensus sequences

+ Create new

« ‹ 1 2 › »

✕ Cancel | ✓ Select

Tools

Discover Tools

Export Datasets

Show Sections

Export datasets to repositories

Export datasets to repositories (Galaxy Version 0.1.0)

Run Tool

Tool Parameters

What would you like to export?

Dataset Collection using their name, structure and extension

Choose your dataset collection *

114: Flexynesis on data 3, data 4, and others: results

Dataset collection will be exported. Collections will become folders and datasets will be files

Include the collection name as top level directory?

No

! Please provide a value for this option.

Directory URI * required

Select

Include metadata files in export?

Yes

Examples of metadata files are bam, cram and vcf indexes that can be regenerated from raw data

Select which characters should be stripped from the exported paths *

Sanitize /

search

Label	Time
Test Dataset for UCSC mafGene	-
Test Datasets for oatk tool	-
HeFDI Data School: Einführung in Galaxy, die Open-Source-Plattform zur FAIRen Datenanalyse	-
Einführung in Galaxy, die Open-Source-Plattform zur FAIRen Datenanalyse	-
invocation export	-
VPT Galaxy wrapper test-data + spatialdata	-
Test data for Sourmash tools	-
Test data for dotPrep tool	-
snapatac2 test-data	-
Data Management Plan	-
Dissemination and exploitation plan	-
test	-
Galaxy Dataset for longdust tool	-
EuroScienceGateway MS6: Integrated EuroScienceGateway knowledge graph	-
WorkflowHub Knowledge Graph	-
MuonGalaxy - tools, workflows and more for muons and materials science	-

Back « ‹ 1 2 3 › » Cancel Select

Usable in workflows!

Galaxy Europe

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User Preferences / Galaxy Storage / Create New

CloudFlare R2

This template can be used to connect to your CloudFlare R2 storage. To use these templates you will need to generate CloudFlare R2 access tokens. Following that tutorial, you should have an "Account ID", and "Access Key ID", and a "Secret Access Key".

Select

iRODS Storage

If you have an account on an iRODS server, you can use such an object store as a storage location for your Galaxy datasets; they will be stored in the iRODS of your choice.

Select

Onedata Storage

If you have an account in Onedata, you can use such an object store as a storage location for your Galaxy datasets; they will be stored in the Onedata space of your choice.
The minimal supported Onezone version is 21.02.4. More information on Onedata can be found on Onedata's website.

Select

Amazon Web Services S3 Storage

Amazon's Simple Storage Service (S3) is Amazon's primary cloud storage service. More information on S3 can be found in Amazon's documentation.

Select

Azure Blob Storage

Setup Azure Blob Storage to enable it as Galaxy storage backend.

Select

Google Cloud Storage

Connect your Google Cloud Storage to enable it as Galaxy storage backend. For the setup you will need to generate HMAC Keys - these can be linked to your user or a service account. Additionally, you will need to define a default Google cloud project to allow Galaxy to access your Google Cloud Storage via the interfaces described in this formular.

Select

Any S3 Compatible Storage

The APIs used to connect to Amazon's S3 (Simple Storage Service) have become something of an unofficial standard for cloud storage across a variety of vendors and services. Many vendors offer storage APIs compatible with S3. Here, you can configure such service as a Galaxy storage as long as you are able to find the connection details and have the relevant credentials.

Select

Rucio Storage

Rucio manages large volumes of data spread across facilities at multiple institutions and organisations. More information on Rucio can be found in Rucio's official site. Currently supported only userpass authentication type.

Select

Workflow: Variant calling and consensus construction from paired end short read data of non-segmented viral genomes v0.2 (Version: 2)

Run Workflow

Send results to a new history | Attempt to re-use jobs with identical parameters? | Send outputs and intermediate to different Galaxy storage? | Primary Storage | Expanded workflow form →

Preferred Target Galaxy Storage

No selection has been made for this workflow invocation. Defaults from history, user, or Galaxy will be used.
Change Preferred Galaxy storage by clicking on the storage button in the workflow run header.

edited 27 days ago

Peter van Heusden | Wolfgang Maier | workflow runs: 2

Variant calling and consensus sequence generation for batches of Illumina PE sequenced

ARTIC | virology

Paired collection of sequencing data *

The PE sequencing data for your batch of samples organized as a list of pairs collection

16: test data

Reference annotation *

Needs to be provided in GTF format. Please make sure it is matching your reference genome sequence!

1: measles.gtf

Fasta reference genome *

A reference sequence for your virus

2: measles.fa

Invocation Preferred Galaxy Storage

The default Galaxy storage for the outputs of this workflow invocation

Use Defaults

If the history has a preference set, that will be used. If instead, you've set an option in your user preferences - that will be assumed to be your default selection. Finally, the Galaxy configuration will be used.

Short term storage for e.g. method development

S3-based object storage is maintained by the compute center of the University of Freiburg. Data on this storage, also called scratch storage, is deleted if it is older than two months (60 days), so it is only appropriate for short-term methods development. The rapid deletion of stored data enables us to provide this storage with an extended quota. This storage is not backed up. More information about our storage offerings is documented at https://galaxyproject.org/eu/storage/.

14.4 GB of 2 TB used

0% of disk quota used

Select as the preferred one

Unshareable storage with additional safeguards

S3-based object storage is maintained by the compute center of the University of Freiburg. Data on this storage cannot be made public, cannot be shared between users, etc.. All your data in Galaxy is by default only available to you and can not be seen by other users. On other storage classes, you can normally always share data, and histories with others. However, this storage prevents sharing and provides an additional safeguard for you and your data. More information about our storage offerings is documented at https://galaxyproject.org/eu/storage/.

27.6 TB used

Select as the preferred one

University Tenant

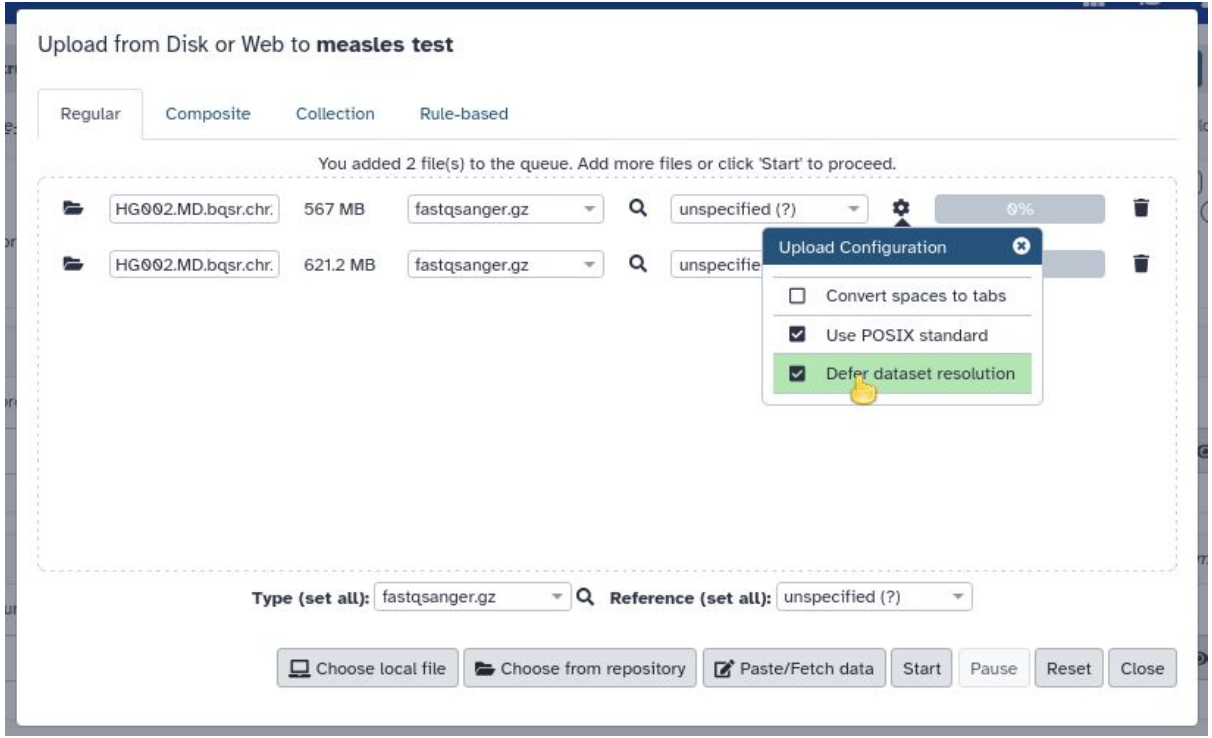
Select as the preferred one

Select as the preferred one

Close

Your own storage doesn't count towards your user quota & you stay in full control of your data

Combine with “Deferred dataset” functionality
and Galaxy won’t store your input datasets either.



Thank you for your attention!

- Galaxy Hub with links out to almost everything else: https://galaxyproject.org/
- Galaxy technical documentation: https://docs.galaxyproject.org
- Galaxy training material: https://training.galaxyproject.org
- Galaxy help forum: https://help.galaxyproject.org/
- Galaxy Europe: https://usegalaxy.eu

https://mstdn.science/@galaxyproject

https://bawü.social/@galaxyfreiburg